

Autonomous Multi-Agent Exploration of Unknown Terrain Using Heterogeneous Ground and Aerial Robots

H-SLAM Research Proposal

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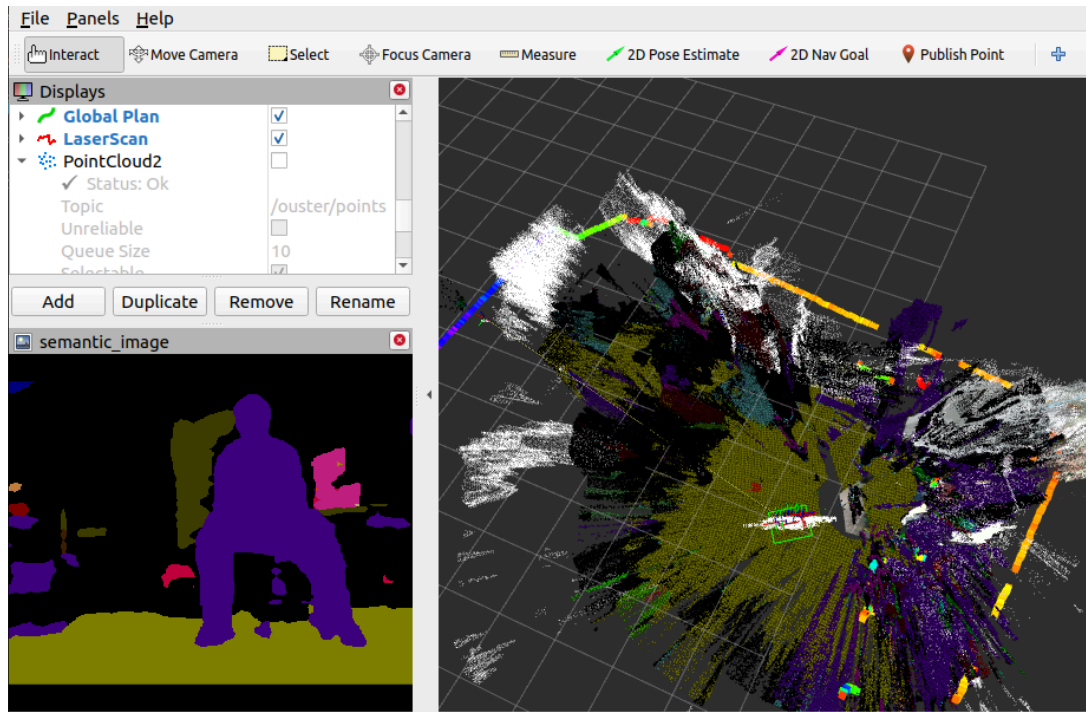


Background

Indoor search-and-exploration tasks increasingly call for heterogeneous robot teams pairing the endurance and payload capacity of ground vehicles with the reach of small aerial platforms. However, most semantic mapping and multi-agent planning methods are validated only in simulation or on single-platform systems, and real-world deployments often depend on external motion capture infrastructure, limiting scalability. We hypothesize that a fully self-contained, infrastructure-free heterogeneous swarm could be built and deployed on real hardware to close this gap. We integrated a Clearpath Jackal ground vehicle with four Bitcraze Crazyflie 2.0 quadcopters, each equipped with a flow deck v2 for onboard localization, removing any need for external tracking. As the Jackal traverses an uncharted room, it builds a real-time 3D semantic map using an existing ORB-SLAM2 and PSPNet-based segmentation pipeline to detect objects of interest. Crazyflies launch from a mounting platform on the Jackal and are routed to detected goals using a planner built on a Graph of Convex Sets (GCS) framework to compute collision-free trajectories for all quadcopters simultaneously in contrast to a purely reactive scheme that dispatches each Crazyflie individually as targets are found. This requires integrating mechanical and programmatic launch of four Crazyflies from a moving platform, formulating convex regions from the semantic map, and linking detections to flight paths. Preliminary trials confirm successful launch, flight, and navigation to detected objects using the Jackal's semantic map, with GCS-based planning under active development. These results indicate infrastructure-free heterogeneous ground-air swarms are a feasible approach for real-world semantic exploration, with GCS planning expected to further improve coordination.

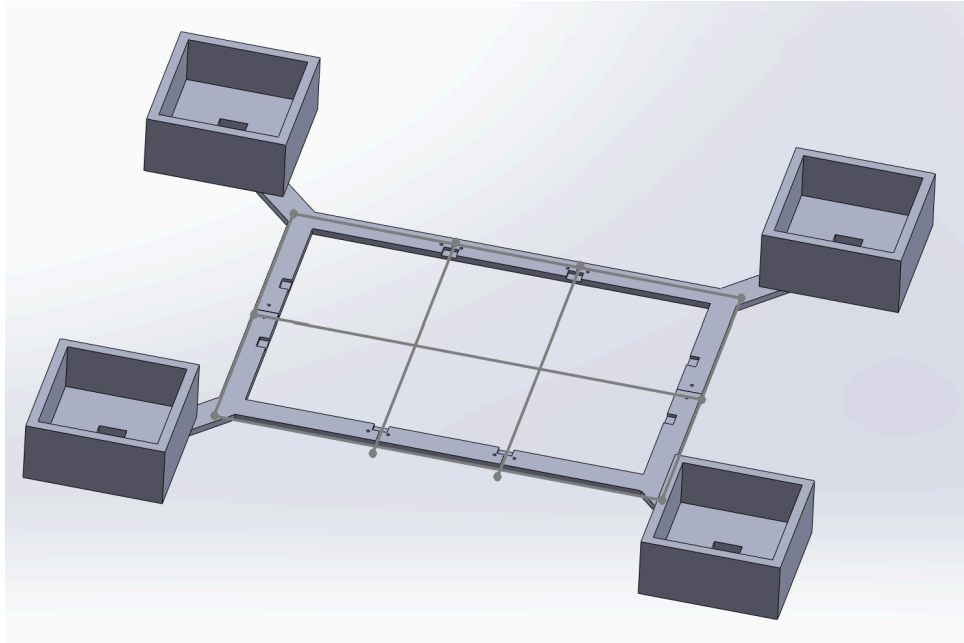
Proposition

Foundations



This past semester, we worked on bringing the Semantic SLAM¹ algorithm to life using the Clearpath Jackal through simulation and reality. Now, our goal is to actually implement the Jackal in a multiagent system where other robots interact with the Jackal's Semantic Map. When we were first considering what other robot we should pair with our Jackal, we naturally landed on the CrazyFlies given that our lab hasn't worked with distributed ground and aerial networks that much and the vast applications of such a project. For instance, this could be applied in the real world through search and rescue missions, planetary exploration, or any other unmanned terrain that comes to mind. Currently our plan is to use a set of 4 Crazyflies with CrazyFlow odometry decks to complete autonomous navigation missions based on our semantic map.

Mechanics



We've designed and are currently in the process of building a carrier to put on top of the Jackal so that as the Jackal navigates around its area, we can input navigation goals through *RViz* for the Crazyflies to launch from the carrier and fly to their navigation goals. The carrier mechanism works by placing the frame on top of the Jackal, and having 4 pods extend diagonally out of the frame. Each pod can hold a single Crazyflie and contains a square hole on the bottom for the Flow Deck to read its coordinates relative to the ground. To ensure that the drones are kept secure, we have a lid on top of each pod which opens with an SG90 servo right before deployment.

Electronics

Of course, these servos need to receive signal and power in order for the lid mechanism to work, so we've attached a buck converter to the Jackal Rover's power distribution board (12V $\rightarrow$ 5V) which then connects to our ESP32 microcontroller containing our servo power board which holds our 4 servos. The microcontroller is then connected via a USB Serial port on the Jackal so that we can communicate to it through our host computer. By doing this, we don't rely on external power for our servos, and can instead use the Jackal's battery as a single power source which we can also communicate with by turning the servo signals into a ROS node. This has been tested without the frame, and we've found the servos to respond to commands effectively through a ROS script on our host computer.

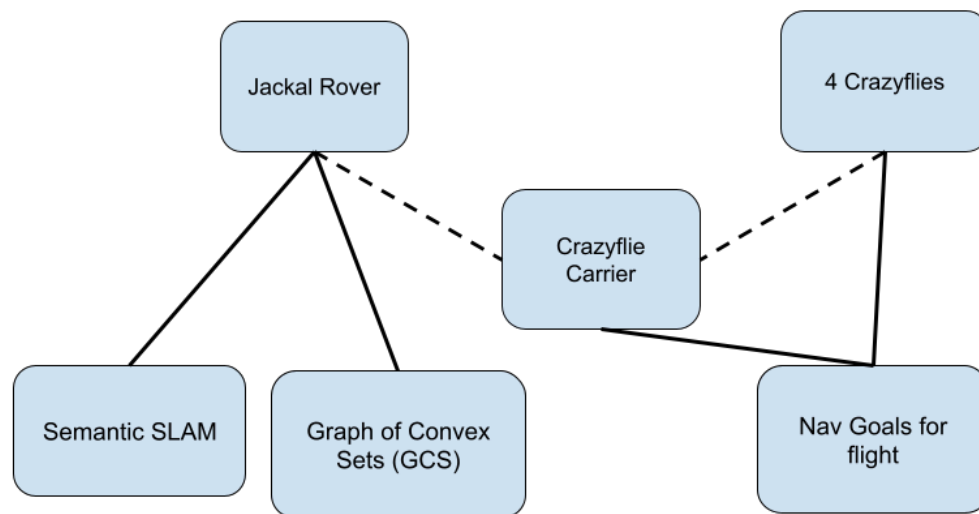
Software

Apart from the embedded firmware on our microcontroller for the carrier. We've also been able to command our 4 Crazyflies to fly to navigation goals via *RViz* and be able to visualize their

positions and orientations pretty accurately. The one setback is that, while the drones land pretty close to their goals, it's not always very accurate due to accumulated drift on the Flow Decks. Because of this, we can do multiple flights with the Crazyflies and have them land generally where they're supposed to, but we can't have them return to their respective pods (or land on a Turtlebot which was an idea pitched to us).

In terms of the Jackal, we've been considering alternate path planning algorithms that could go with our Semantic SLAM algorithm. We met with *Shilin You*, a graduate research assistant working with *Dr. Tyler Summers* and learned about GCS (Graph of Convex Sets)² which would essentially have our rover traverse numerous points on curved paths before reaching a target. This immediately stood out to us as it would give our rover the ability to autonomously, semantically map its environment with zero human input whatsoever, and then allow the Crazyflies to traverse the area aerially to visit points of interest in the map.

Systems Architecture



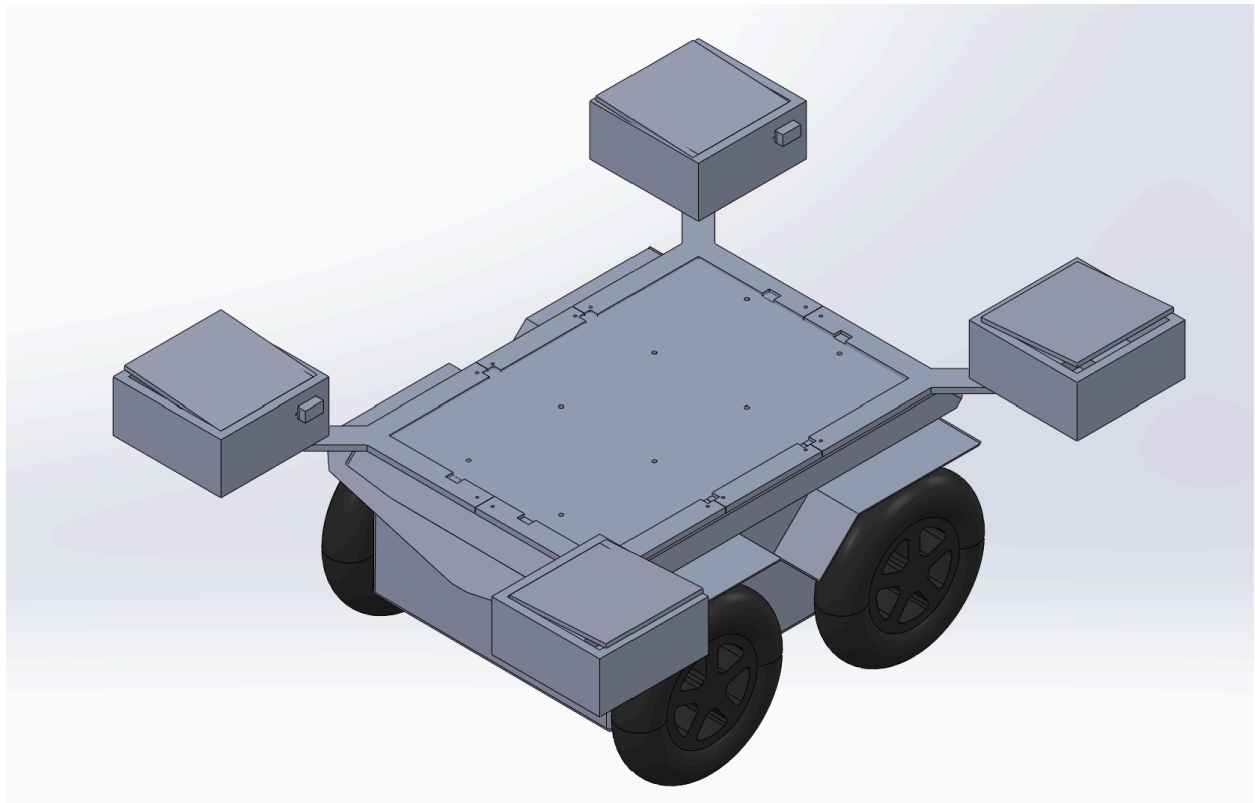
Given the complexity of any multiagent system, we've created a graph to show how simple our setup really is. The Jackal Rover maps and traverses the general area while having an attached Crazyfly carrier, each pod holding a single drone. From there, the goal of the Crazyflies is to navigate the area, visiting points of interest in the Semantic Map.

To achieve this, the system relies on a tightly coupled hardware and software pipeline. The Jackal Rover first utilizes ORB-SLAM2 and PSPNet to construct the real-time 3D semantic map. Once target waypoints are identified, the Graph of Convex Sets (GCS) path planner on the Jackal generates collision-free trajectories. Programmatically, the Jackal then communicates

over a USB serial connection to an onboard ESP32 microcontroller, triggering SG90 servos to open the lids of the Crazyflie Carrier.

Upon release, the Crazyflies leverage their onboard Flow Deck sensors to handle local positioning completely infrastructure-free, bypassing the need for external motion capture systems. This centralized planning and decentralized execution loop allows the heterogeneous team to cooperatively explore and analyze unknown environments with high autonomy.

Progress



Jackal Mapping and Navigation

So far, we've got Semantic SLAM and the RRT* path planning algorithm working on the Jackal. We had issues in the past where the rover wasn't able to detect obstacles which we figured out to be due to the *Ouster* Lidar running on a different clock then the rest of the rover keeping it almost 30 seconds behind real time, which was very visible in *RViz* as we noticed a lot of Pointcloud lag. We fixed this by first going through the Ouster Hardware Setup Manual to get lag down to a 2 second difference and then hacking into the lidar a bit to adjust our resolution and refresh rate.

Crazyflie Navigation

Our 4 Crazyflies are able to fly to arbitrary waypoints set in RViz and visualize their coordinates with pretty strong accuracy with sole reliance on Flow Deck odometry. Given that we have no reliance on a motion capture system, our Crazyflies have the ability to fly in any environment. By doing this, we're able to have our multiagent system (rover, carrier, and drones) function in any area within or outside of the lab to eventually reach our goal of mapping and navigating the environment they run in. We've also attached stilts on our Crazyflies to keep them from getting bad readings from the Flow Deck when it's on ground. In addition to this, we've been working on having the Crazyflies run in curved paths themselves to allow them to fly at the same time without running into each other as they would if they ran straight paths to the same waypoint.

Todos

As we've recently completed our Carrier CAD, we need to print and assemble our carrier to place it on top of the Jackal and launch our Crazyflies from the carrier. We've been prototyping with pods and have found that, despite the fact that we don't use a motion capture system, keeping the mocap frames on the Crazyflies keeps them steady in the pods, ensuring that they don't move around in the pods as the Jackal moves about. At this point, we're done prototyping and have tested flying out of the pods and have found our current pod model to be very efficient for launches.

In addition to this, we also want to implement the GCS path planner on the Jackal to fully allow our rover to autonomously navigate and map its entire environment before deploying Crazyflies. By doing this, we'd no longer have to set nav goals around our environment to have the rover map everything it sees, and instead have waypoints the Jackal traverses to on its own before reaching its target and having the general area mapped.

Future Goals

Currently, our project has the Jackal map out its area and the Crazyflies simply going to points in that map. With this setup, the Crazyflies aren't really making a valuable contribution to the multiagent system. Thus, we want to later explore attaching cameras to our Crazyflies along with their existing Flow Decks so that the Crazyflies can actually expand the Semantic Map and traverse areas the Jackal can't reach on ground. We've already come across some very strong papers online that already cover attaching high FOV (Field of View) cameras on the Crazyflie 2.0 without having to change batteries or decreasing flight time. One that specifically caught our attention was a paper done on vision based control in launching and landing the Crazyflie 2.0³ whose onboard hardware matched ours quite well. Thus, we'll explore these options when the time comes to it in order to create a highly valuable research project.

Citations

[1] J. B. M. Lagria, "Autonomous Navigation using Active Visual-Semantic Slam." Order No. 31963567, The University of Texas at Dallas, United States -- Texas, 2024.

[2] S. You et al., "Motion Planning with Precedence Specifications via Augmented Graphs of Convex Sets," Dept. Mech. Eng. and Dept. Comput. Sci., Univ. Texas at Dallas, Richardson, TX, USA, Tech. Rep., 2026.

[3] C. Karlsson, "Vision based control and landing of Micro aerial vehicles," Bachelor's thesis, Dept. Eng. Phys., Karlstad University, Karlstad, Sweden, 2019. [Online]. Available: DiVA, <http://urn.kb.se/resolve?urn=urn:nbn:se:kau:diva-73225>